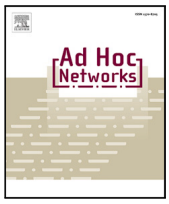
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A context-aware zero trust-based hybrid approach to IoT-based self-driving vehicles security

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ABSTRACT

With the speedy progression and adoption of IoT devices in modern self-driving vehicles (SDVs), autonomous driving industry is gradually reforming its capabilities to provide better transportation services. However, this domain faces enormous security and privacy challenges and thus has become an attractive target for attackers due to its rapid growth and market worth. Furthermore, the rapid transformation in technological tools in transport industry and speedy evolution of cyber-attacks paved the way for designing efficient IDSs. Motivated by these challenges, we put forward a new secure and efficient IDS approach for the security of SDVs. The propose approach utilizes an emerging strategy to mitigate security vulnerabilities and cyber attacks detection using zero trust (ZT) model. Through this work, we put forward a context-aware zero trust security framework for IoT-based SDVs. The proposed framework utilizes a context-aware design to evaluate the trustworthiness of the devices using multi-source trust and reputation model. Then, to make the framework more effective and reliable, we introduce crawler system into the context of the IoT-devices in SDVs to make the system unbiased. Additionally, an observer module is developed that employs state-of-the-art machine learning algorithm to detect malicious actions. Empirical results on two standard benchmark datasets (i.e., Car Hacking and ToN_IoT) validate the practicality and robustness of propose framework in real-world transport systems with enhanced security and trust management against evolving cyber-threats. Detection results demonstrate that the proposed framework secured the best performance by achieving 99.43% and 99.52% accuracy for Car Hacking and ToN_IoT, respectively. The findings of this study will help the security professionals and researchers to comprehend the importance of ZT architecture in developing effective and robust security solutions for modern IoT-based SDVs.

1. Introduction

Self-driving vehicles (SDV) or autonomous vehicles (AV) are the automobiles which does not involve human driver. These vehicles are also termed as driver-less vehicles (DV) and are provided with numerous kinds of software, an efficient processing system having numerous sensors and actuators. Even though an ideal SDV still does not exist in real-world environments, the market for partly DVs or SDVs with several ranks of automated features is highly desirable. It appears that the effective employment of SDVs will enhance the overall economy and transportation networks of the world. Theoretically, it will deliver more protection and safety, and also decrease the rate of accidents, as each year countless accidents happen only because of errors made by the human driver. Moreover, the implementation of an SDV could be reflected as a benevolence service for special people with some disabilities. To achieve this vision, numerous companies are functioning on the successful execution of technological tools for the progression of SDVs. Soon people from different parts of the world will experience the utmost attributes of intelligent transport services through these developments in the forthcoming years.

However, this domain faces significant concern for SDVs such as security and privacy against evolving cyber-attacks. Especially, the attack surface becomes vast when SDVs are equipped with Internet of Things (IoT)-based devices [1]. Since, the SDVs is desirable to entirely rest on communication equipment such as, sensors, external indicators

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for proper working, one could anticipate that this has the potential to attract more hackers and attackers as compared to other vehicles. The effect of both physical and cyber-attacks implemented by invaders can comprise the leak of driver or vehicle information, such as, the number of persons in the vehicle, the gender, the location, etc. To deliver security and safety, it is vital to institute a robust defense scheme should provide protection to both in-vehicle communication (IVC) and vehicle-to-vehicle communication (VtVC). For an SDV, both the IVC and VtVCs are coordinated by the ECU,[1] which are known as the core part of the SDV and are accountable for taking decisions in real-time. Therefore, it is imperative that the ECUs of an SDV exchange data between themselves and other vehicles through a protected communication system.

1.1. Security challenges of autonomous vehicles

For IVCs network, CAN[2] tool is reflected as de-facto amid the vehicle manufacturers and inherent methods [2,3]. But, CAN procedure still suffers from some serious security threats, which can harm the development of intelligent transportation systems (ITS) and can create dangers for people's lives. In reality, this CAN protocol operates in the same way to a distribution system, where there is no system to check the senders' identity. Numerous researchers and scholars have attempt to deliver solutions to enhance the security mechanism of CAN protocol [4–7], but most of the methods work for a specific category of attack. At present, the growing extent of research efforts on SDVs has motivated us to put forward an effort that successfully identify attacks in both IVCs and their external networks.

AVs are termed as rolling machines/computers due to their large dependency on hardware and software systems for proper operation and functioning, internet connectivity, and complex networks. This dependency makes them susceptible to security flaws and different types of cyber-attacks that includes:

- **Spoofing Attacks:** These attacks tries to trick the vehicle into different location or situation.
- **Replay and Fuzzing Attacks:** The Replay attacks tries to gather normal traffic data at a particular time for injection purposes in later times. While, the Fuzzing attacks involves injection of random messages to cause undesirable effects in vehicle's behavior.
- **Tampering Attacks:** These attacks tries to trick the vision of vehicle with false image projections on the road. Furthermore, tampering attacks on software updates can result in gaining unauthorized access to vehicle's control.
- **Data breach Attacks:** These attacks tries to steal sensitive data such as location of vehicle or driver's information.
- **Flooding Attacks:** These attacks tries to exploit the bandwidth of CAN bus through the transmission of large of messages.

Most of the common attacks on CAN-based SDVs include eavesdropping, brute force, replay, DDoS and DoS attacks and route modification. All these susceptibilities and security threats demands for a robust defense mechanism that can prevent SDVs and their networks from probable outbreaks that pose dangers to IVC and VtVC in SDV networks. IDSs[3] are an operative method of security to recognize anomalous data and attacks over the network data throughout the communication among other devices in the internet of vehicles (IoV). Intrusion discovery is habitually deliberated as a classification task; deep learning and machine learning approaches have been broadly applied to build IDSs [8]. This effort put forward an intelligent IDS model to detect attacks not only from CAN-based SDVs but also from exterior networks of SDVs and IoVs.

[1] Electronic Control Unit.
[2] Controller Area Network.
[3] Intrusion Detection System.

1.2. Motivation and contribution

Unlike other security methods, the zero-trust (ZT) security approach trails the principle of ***'verify and never trust'*** and presumes that any accessibility action to a service or resource performed by users or devices both within and outside the system is untrusted and needs to be continuously verified. This ZT-based security approach is a promising method of revolutionizing security of IoT networks without restraining to the range of IoT system. In simple words, the ZT model works in a way that no entity in a system is explicitly trusted. Which means that for every individual entity's action performed (whether is request for a service or try to access any critical data), the network must first verify the authenticity and trustworthiness of the entity before it is granted the clearance to access the resource or data. The current cyber-attack landscape has led us to develop a ZT-based security framework using AI and ML[4]-based methods. Such as grouping (ZT+ML) of these two methods has the potential to revolutionize the modern security tools, for example a ZT-based ML framework will offer more powerful and granular control the security personals of the IoT network.

To fill this void, this study offers a ML-enabled context-aware ZT framework for the security of IoT-based SDVs. The major contributions are summarized as follows:

- We propose a context-aware trust evaluation model using multi-source data to evaluate trustworthiness.
- We develop a crawler method to make the framework reliable, more effective and unbiased.
- We introduce an observer module using machine learning algorithm to identify abnormal data and attacks.

1.3. Organization

The remainder of this article is organized as follows. We overview the existing work related to zero-trust and security of IoT and SDVs in Section 2. The detailed architectural details of proposed framework are presented in Section 3. Section 4 is devoted to the experimental and implementation details and converses the results. We then conclude the article by presenting conclusion and future work in Section 5.

2. Literature study

ML-enabled security methods are influential protection methods and used by many researchers [23–25]. Even though, the traditional security tools are capable to defend IoVs and SDVs from evolving cyber-threats, there still exists security and privacy challenges related to the complexity of IoVs and accurate detection of attacks based on trust architectures [26,27].

There has been a substantial quantity of studies on the security of CAN-based networks due to their prevalent use [4–7,28–32]. Scholars integrate traditional validation procedures to detect illegal CAN ID [33–38]. To detect anomalies in SDV networks, scholars develop a normal profile either considering CANs' internal communication or the specifications of vehicle. But, this system can be straightforwardly exploited by means of present fabrication outbreaks. The writers of [28] developed a detection system using decision trees by considering the cyber and physical attributes, and used a small-scale robotic-based vehicle to authenticate their proposed approach; nonetheless, their system has detection rate with high latency. An alternate exciting study is proposed by [33] which introduced an IDS based on FPGA to detect attacks in vehicular networks. However, this technique is only applied for the platforms that uses FPGA, which confines its deployment to other systems or networks that are GPU or CPU-based.

[4] Machine Learning.

Table 1
Summary of existing work on IoT and self-driving vehicles.

Ref.	Domain	Model	Strengths	Limitations
[9]	IoT Network	Ensemble learning	Good performance in accuracy and detection rate	Non-trust-based approach and high FAR
[10]	IoT Network	Convolutional neural network (CNN)	Good performance in accuracy	Not suitable for large batch size, and Non-trust-based approach
[11]	IoT Network	Long short-term memory model (LSTM)	Simple structure and good performance at back-end level	Not suitable for detection of new attacks, and Non-trust-based approach
[12]	IoT Network	Bidirectional Long Short-Term Memory (BiLSTM)	Blockchain-based model with good accuracy rate	performance degrades when handling large quantity of data, not suitable for detecting complex threats
[13]	IoT Network	Long short-term memory model (LSTM), and also Random Forest (RF)	Enhanced F-score performance	Tested with only one dataset, and Non-trust-based approach
[14]	IoT Network	Deep Neural Network (DNN)	Tested with four datasets, and achieved good accuracy performance	Tested with only four families of attacks, and Non-trust-based approach
[15]	IoMT Network	Deep Neural Network (DNN)	Good performance with improved accuracy rate	Tested with non IoT/IoMT data, and Non-trust-based approach
[16]	IoMT Network	Ensemble learning	Good detection rate and accuracy rate	High FAR, and Non-trust-based approach
[17]	IoT Network	Long short-term memory model (LSTM)	Achieved good performance using F-score and detection rate	Require huge training data, and takes longer time to train
[18]	IoMT Network	Blockchain and Deep Learning	Achieved higher accuracy rate	Tested with non-IoT datasets, and Non-trust-based approach
[19]	IoT Network	Feed-forward neural networks (FNN)	Achieved good performance in multi-class classification	Low performance against various attacks such as information theft and keylogging
[20]	Vehicular Networks	Generative adversarial networks (GANs)	Results shows 16.4% reduction in attacks	Require a massive amount of human effort in the pre-processing stage
[21]	Connected and Autonomous Vehicles	Zero-trust architecture	Simple architecture and easy to use	Only focuses on CAN bus communication, and not suitable for IoT-based network
[22]	Autonomous Vehicles	Zero-trust architecture	Improved performance	Only focuses on data reconstruction attack

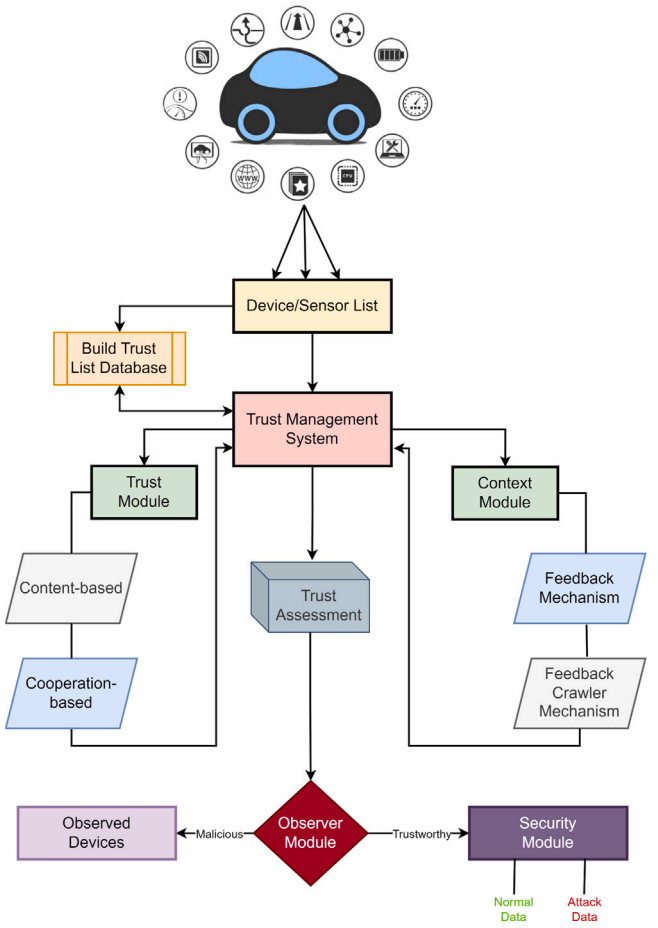


Fig. 1. Flowchart of the proposed zero trust security framework.

The writers of [4,36] developed a system for SDVS by utilizing the CANs' message periodicity. Practically, ECUs creates CAN communication at a particular frequency. Consequently, when an exterior invader injects messages, it is likely to identify abnormalities in inter-arrival time. Another effort by [34] proposed one-class SVM-based to detect anomalies in IoVs on the basis of frequency. Although, their proposed method has high accuracy rate, but actual CAN messages are variation prone, and often reveals varying time of inter-arrival time, which decreases the trustworthiness of these proposed methods.

Other scholars proposed IDS methods based on entropy to detect attacks in CAN-based networks [35,36]. The fundamental notion of these methods is that these systems will have stable or standard entropy for attack-free CAN messages. In contrast, the abnormal messages of CAN will have unsteady behavior pattern. The authors of [38] proposed a method based on sequencing of arbitration ID. This technique forms a transition matrix by means of CAN communication and equates abnormal CAN communication ID pattern to identify an outbreak. Although, their proposed technique can identify simple mimicry outbreaks; but, it is unable to other attacks, such as replay attacks.

The existing literature is summarized in Table 1 by highlighting their strengths and limitations. As seen from Table 1, most of the existing work focuses more on performance factor hence the zero-trust aspect is neglected. Further, IoT-based zero-trust architecture is least studied by the research community due to non-availability of realistic representation of these networks. Therefore, this work tries to fill these gaps by offering a zero-trust-based intelligent security framework for the protection of IoT-enabled SDVs.

3. Proposed ML-enabled context-aware zero trust framework

3.1. System overview

The proposed framework is a layered design as shown in Fig. 1. The primary layer consists of SDVs equipped with IoT-based devices that sense and generate data which is then transmitted to the device list layer. After receiving the data, this layer perform some preprocessing steps (such as label encoding and scaling) and compute the trust scores (utilizing both content-based and cooperation-based trusts) for the IoT devices and sensors using the developed trust management system. The context of the devices/sensors are then updated through feedback and feedback crawling mechanisms, where less ranking sensors are monitored by observation module. The sensors/devices are then evaluated for their trustworthiness of the participating IoT devices. Finally, the last layer receive the data and detect the intrusions and abnormalities using ML model. The Pseudocode of these steps are presented in Algorithm 1. It is worth mentioning that modern network devices and SDVs are capable of deploying security softwares as they are equipped with adequate amount of memory and computing resources.

Algorithm 1: Proposed ML-enabled Zero-Trust Security Framework

Require: Read Input Data and Initialize parameters
Initialization:
Dataset ← Label Encoding;
Dataset ← Scaling/Normalization;
Load $\mathcal{TL}^{SDV_{sid}}$ ← Using Eq. (1);
Set ← *Trust-Threshold*;
Set ValueSets for ← α, β, γ (weight factors);
Context-Aware Trust Module:
for ∀ $\mathcal{TS}_{SDV_{sid}}$ **do**
  Compute context process using Eq. (3);
  Compute ranking process using Algo. (2);
  Compute feedback and crawling mechanisms using Eq. (5) and Eq. (6);
  Compute $\mathcal{OM}_{SDV_{sid}}$ using Eq. (7);
end
Security Module:
TrainingData ← Dataset[$Training - Instances$];
TestingData ← Dataset[$Testing - Instances$];
for ∀ *Dataset[Training+Testing]* **do**
  Train and predict using Eq. (10);
  Compute the objective function using Eq. (11);
  Control over-fitting using Eq. (12);
  Calculate loss using Eq. (13);
  Optimize the objective function using Eq. (14);
end
Result: Print and visualize the detection results

3.2. Trust module

The writers of [39] defined trust as "*Trust is the extent to which one party is willing to depend on something or somebody in a given situation with a feeling of relative security, even though negative consequences are possible*". Similarly, the authors of [40] describe it as "*Trust is the subjective probability by which an individual, A, expects that another individual, B, performs a given action on which its welfare depends*". In terms of IoT environments, the authors of [41] express trust as "*A Thing's belief in another Thing's honesty, reliability, and capabilities recommended by other Things*". By looking at these several definitions, we can realize that trust can be of different types such as: Direct; Derived; and Recommended Trust.

The major component of the proposed design are presented as follows:

SDV denotes a SDV in our case and sid represents the unique identification number for a particular IoT sensor. These sensors generate and transmit data to the collection point denoted by DCP which is then forwarded to the decision engine $D\mathcal{E}$ for further processing. The assessment of trust scores ($\mathcal{TS}$) depends on the public key for particular SDV. For simplification purposes, we assume SDV_{sid} focused method to identify a unique SDV for the trust assessment ($\mathcal{TA}$) process during $\mathcal{TS}$, where the range of $\mathcal{TS}$ is [0, 1]. The $\mathcal{TA}$ (trust assessment) part consists of two major elements i.e., A_1 and A_2, where A_1 carries all the values to assess the trustworthiness of SDV_{sid} while A_2 contains the repository having $\mathcal{TS}$ of all SDVs $\sum_{j=1}^{m} \mathcal{TS}_{SDV_{sid}}$. In the $\mathcal{TA}$, the $\mathcal{TS}_{SDV_{sid}^{DE_j}}$ is forwarded to $D\mathcal{E}$ to calculate the trustworthiness. This trust calculation process also involves an incentive/penalty mechanism which is based on data contents. Therefore, the device/sensor participation in the trust calculation process will result in gain or loss in trust scores. For instance, the gain or loss can be computed using the context-aware feedback $\mathcal{CAF}$ for a specific sensor $\mathcal{CAF}_{SDV_{sid}^{DE_j}}$ where $\mathcal{CAF} \in [-1, 1]$. To prevent security issues in the trust management system, we also developed an observer component in the system to distinguish malicious devices to assist the framework in detecting trusted and untrusted devices.

The $\mathcal{TA}$ process involves two techniques i.e., Content-based and Cooperation-based methods which are briefly defined below:

- **Content-based:** This method is based on the contents it transmits or sends. The $D\mathcal{E}$ computes the relevance of context ($\mathcal{CON}$) between connecting devices/sensors and calculate the gain/loss in trust score using the incentive/penalty system. This method not only assists in detecting malicious data send by the device/sensor in the network but also helps to control the cold-star issue.
- **Cooperation-based:** Using this method, the $\mathcal{TS}_{SDV_{sid}}$ of an SDV is build by associating it with the $\mathcal{CON}$ of other sensors/SDVs. This method helps the framework to identify the current trust score level of a particular device/sensor upon its communication with other sensors. Through this Cooperation-based approach, the system's $D\mathcal{E}$ can effectively determine the trustworthiness of a communicating device and identify its maliciousness using both the interactive experience and its own basis.

For reliable and effective $\mathcal{TA}$, we developed a multi-source trust score method using the $\mathcal{CON}$ and reputation of the participating SDV. In this method, the framework will collaborate with communicating sensors/SDVs to get involved in the $\mathcal{TA}_{SDV_{sid}}$ process based on their preceding $\mathcal{TS}_{SDV_{sid}}$ context relevance and reputation. Here in this work, only the directly connected devices/SDVs are considered in the $\mathcal{TA}$ process. This step is presented as:

$$\mathcal{TL}^{SDV_{sid}} = \sum_{j=1}^{m} \mathcal{TS}_{SDV_{sid}^{j}} \tag{1}$$

where $\mathcal{TL}$ denotes the trust list which contains trust scores of all other SDVs/sensors that it has interacted previously. However, if no history of interaction is found then by using the content- and cooperation-based methods, the $\mathcal{TA}$ can be represented as:

$$\mathcal{TS}_{SDV_{sid}}^{CS,D\mathcal{E}} = \beta \times \sum_{j=1}^{m} \frac{\mathcal{TS}_{SDV_{sid}}^{CS_j^{TL}}}{M} + \gamma \times \sum_{j=1}^{m} \frac{\mathcal{TS}_{SDV_{sid}}^{D\mathcal{E}_j^{TL}}}{M} \tag{2}$$

the weight factors γ and β can be tuned appropriately. Hence, the entire $\mathcal{TA}$ process can be shown as:

$$\mathcal{TS}_{SDV_{sid}}^{New} = \alpha \times \mathcal{TS}_{SDV_{sid}}^{Current} + \beta \times \sum_{j=1}^{m} \frac{\mathcal{TS}_{SDV_{sid}}^{CS_j^{TL}}}{M} + \gamma \times \sum_{j=1}^{m} \frac{\mathcal{TS}_{SDV_{sid}}^{D\mathcal{E}_j^{TL}}}{M} \tag{3}$$

This $\mathcal{TA}$ process is ranked through a ranking protocol that uses context of the sensors/SDVs. This ranking method is designed to develop an evaluation system using the assistance of neighboring sensors that

Table 2
Ranking scores and interpretations.

Score	Interpretation
0 - 0.3	Bad/Low Ranking
0.4 - 0.6	Average Ranking
0.7 - 1	Good Ranking

are well ranked and trustworthy. The ranking process can be expressed as:

$$\mathcal{RS}^{CON}_{SDV_{sid}} = \sum_{s \in A_{SDV_{sid}}} \frac{R(s)}{AS(s)} \tag{4}$$

where $A_{SDV_{sid}}$ consists of $\mathcal{RS}^{CON}_{SDV_{sid}}$ from every well ranked associated links to the device SDV_{sid}, the ranking of sensor is denoted by $R(s)$ and the aggregated quantity of associations from $A_{SDV_{sid}}$ is denoted by $AS(s)$. These steps are briefly defined in Algorithm 2, and the ranking score with their respective interpretations in Table 2

Algorithm 2: Ranking Process

```
Result: Ranking level/score
Require: SDV_sid
for ∀ RS^CS_SDV_sid AND RS^DE_SDV_sid do
    Use Eq. (4) to calculate RS_SDV_sid;
    if RS_SDV_sid ≥ Threshold
    AND
    RS^CON_SDV_sid = SDV^CON_sid then
        return RS^CON_SDV_sid;
    else
        return 0;
    end
end
```

3.3. Context module

The context module is one of the important part of this work as it involves the context of devices/SDVs. If the record of an SDV or sensor is not available in the trust repository, then the framework will assess its trustworthiness level according to criteria discussed in previous section. Additionally, the this new device/SDV will be observed by the observer component to keep track of its activities. Also to resolve its context, the $\mathcal{DE}$ will compute its $\mathcal{CON}$ for the particular device/SDV using the content-based $\mathcal{TA}$. Alternatively, if the record of an SDV or sensor is available in the trust repository, then the relevance of $\mathcal{CON}$ will be calculated using the loss/gain in the trust values. This context determination module have three basic components such as, a feedback mechanism, a feedback crawler mechanism, and an observer component.

- **Feedback Mechanism:** This component assists in computing the loss/gain in the trustworthiness of devices. After the trust of an SDV or device is assessed, the $\mathcal{DE}$ issues $\mathcal{TS}_{SDV^{\mathcal{CON}}_{sid}}$ to all the devices that participated in the $\mathcal{TA}$ process. This process can help the $\mathcal{DE}$ in other SDV's or networks to compute the $\mathcal{TA}_{SDV^{\mathcal{CON}}_{sid}}$ of future sensors.

$$\begin{aligned} \mathcal{CAF}^{\mathcal{DE}}_{SDV_{sid}} &= \mathcal{TS}^{New}_{SDV_{sid}} - \mathcal{TS}^{Current}_{SDV_{sid}} \\ -1 \geq \mathcal{CAF}_{SDV_{sid}} &\leq 1 \end{aligned} \tag{5}$$

  Here, it is worth mentioning that a negative value represents loss in trust score and positive values denotes gain. This context-aware feedback mechanism ensure reliability and fairness in the $\mathcal{TA}$ process. Similarly, the $\mathcal{CAF}^{\mathcal{DE}}_{SDV_{sid}}$ can also assist in computing loss/gain of sensors based on the worth of their contents. This framework works on the basis of punishment/reward system in $\mathcal{TS}$ to help in identifying the untrustworthy/malicious sensors and keep the framework unbiased.

- **Feedback Crawler Mechanism:** A $\mathcal{TL}$ is generated using the feedback crawler mechanism after a particular amount of time to form extensive $\mathcal{TA}$ process where the $\mathcal{DE}$ reassess $\mathcal{TS}_{SDV_{sid}}$ of specific sensors. This reassessment process is based on trust scores, context, and behavior of a sensor/SDV. Using this approach, the SDVs can sync the $\mathcal{TS}$ of sensors based on their interchanges and is given as:

$$\mathcal{FCM}^{\mathcal{DE}} = \mathcal{TL}_{tp} \tag{6}$$

  where the particular time period is denoted by tp.

- **Observer Component:** This component is designed to add an additional security layer to the proposed framework by observing behaviors of devices and report any suspicious activity. For example, the observer mode ($\mathcal{OM}$) will be activated for a particular device/SDV if the $\mathcal{TS}_{SDV_{sid}} < Trust\text{-}Threshold$ for the $\mathcal{FCM}^{List}_{SDV_{sid}}$. This process can simply be expressed as:

$$\mathcal{OM}_{SDV_{sid}} = \begin{cases} 0, & \text{if } \mathcal{TS}_{SDV_{sid}} > Trust - Threshold \\ 1, & \text{if } \mathcal{TS}_{SDV_{sid}} < Trust - Threshold \end{cases} \tag{7}$$

3.4. ML-based security module

The proposed framework utilizes a ML method named gradient boosting method for an extra level of security. This work implements a scalable ensemble method known as GBM.[5] This ensemble learning technique denotes a systematic architecture that can aggregate several weak learners to form a more comprehensive and powerful model. In this model, decision trees (DT) are generated and every tree represents a classifier/learner. These trees are build in a sequential manner where every successive tree depends on the shortcomings of the previous tree. In this way the tree trains itself using the error updates from its predecessors. We can define this process as:

$$\sum_{n=1}^{\mathcal{N}} f_n \ , \ f_n \in \mathfrak{T} \ , \quad n = 1, 2, 3, \ldots, \mathcal{N} \tag{8}$$

where the number of DTs is denoted by $\mathcal{N}$ and the predictions are defined as f_n. By combining all these trees, we can compute predictions on the input–output pairs $(\overrightarrow{a_i}, b_i)$ of the dataset, where the k-dimensional vector of input $\overrightarrow{a} = \{a_1, a_2, a_3, \ldots, a_i\}$ denotes the sum of all detection results for every tree, and the 1-dimensional output is denoted by b_i that corresponds to $[0, 1]$ (0 for normal and 1 for abnormal). This sum is shown by:

$$\hat{b_i} = \sum_{n=1}^{\mathcal{N}} f_n(a_i) \tag{9}$$

where the vector of feature for ith point of data is denoted by a_i. At every iteration, the model is trained using a tree which is then added to the tree list. Hence, the prediction at any point of time (tth) can be shown as:

$$\hat{b_i}^{(t)} = \sum_{n=1}^{t} f_n(a_i) \tag{10}$$

The two main functions of this method are the regularization function Ω, and the loss function $\mathcal{L}$ where the simplicity and over-fitting is controlled by Ω, and the productivity is controlled by $\mathcal{L}$. Therefore, we can define the objective function as:

$$\text{ObjFun} = \Omega + \mathcal{L} \tag{11}$$

where Ω is computed as:

$$\Omega = \gamma\tau + \frac{1}{2}\lambda \sum_{i=1}^{\tau} \mathbb{Z}_i^2 \tag{12}$$

[5] Gradient Boosted Machine.

and $\mathcal{L}$ is computed as:

$$\mathcal{L} = -\frac{1}{\mathcal{M}} \sum_{j=1}^{\mathcal{M}} b_j \log\left(\hat{b_j}\right) + (1 - b_j) \log\left(1 - \hat{b_j}\right) \tag{13}$$

where the calculated output is denoted by $\hat{b_j}$, the amount of tree leaves is denoted by τ, the degree of regularization is controlled through λ and γ parameters, and the rank on ith leaf is denoted by $\mathbb{Z}_i^2$. By combining Ω and $\mathcal{L}$, we can reformulate the ObjFun as:

$$\begin{aligned} \text{ObjFun}^t = \hat{b_i}^{(t)} &= \sum_{i=1}^{t} \Omega(f_i) + \sum_{i=1}^{\mathcal{M}} \mathcal{L}\left(b_i, \hat{b_i}^{(t)}\right) \\ \hat{b_i}^{(t)} &= \sum_{i=1}^{t} \Omega(f_i) + \sum_{i=1}^{\mathcal{M}} \mathcal{L}\left(b_i, \hat{b_i}^{(t-1)} + f_t(a_i)\right) \end{aligned} \tag{14}$$

where the training of model at t iteration is denoted by $\hat{b_i}^{(t)}$, which is the prediction of tth round to the current prediction $f_t(a_i)$ and earlier round $\hat{b_i}^{(t-1)}$ *a.k.a* boosting method or additive method.

4. Experimental evaluation and results

4.1. Experimental setup

To evaluate the performance of the proposed framework, we conducted extensive experiments on two real-world datasets, i.e., CHADC[6] dataset [42], and ToN_IoT dataset [43]. The framework is implemented in Python-based TensorFlow library architecture using Anaconda. The system specs include a *Ubuntu* operating system having *Intel(R) 4110 @ 2.10GHz* CPU, *NVIDIA*-based *RTX2080* GPU, *1,28e+8 kB* Ram, and Scikit-learn libraries. For the XGB-based security module, the parameters were set as: booster type = gbtree, refresh leaf = 1, maximum depth = 10, number of parallel tree = 1, importance type = gain, learning rate = 1, subsample ratio = 0.5, max delta step = 0, sampling approach = gradient-based, n_estimators = 20, and random_state = 2.

4.2. Evaluation data

The performance of the proposed framework is evaluated using two public datasets since they realistically represents the contemporary IoT and SDV environments. A brief details of these datasets are as follows:

4.2.1. CHADC data

This dataset was created by [42] in Cybersecurity competitions which involved both detection approaches and attack on CAN-based an in-vehicle network. The vehicle used in the generation process was a car model released in 2000 *Hyundai Avante CN7*. The challenge in this competition was aimed to both attacks and detection algorithms concurrently. The CAN traffic was captured during the process to helped security researchers for developing and testing forthcoming security approaches. This dataset contain both normal data and attack data that include four types of attacks such as Fuzzing, Spoofing, Flooding, and Replay attacks. The dataset include 6 fields as outlined in Fig. 2.

4.2.2. ToN_IoT data

This dataset is created by cybersecurity team [43] using IoT testbed design. The testbed involved several heterogeneous kinds of IoT devices such as Thermostat, Fridge, GPS, Motion Light, *etc*. This dataset contains seven attack families as outlined in Table 3. This dataset contains 22 fields and has labeled normal and abnormal records.

4.3. Performance evaluation and results

The performance of the proposed framework is evaluated using the datasets mentioned in Section 4.2. To properly evaluate the proposed framework, we conducted experiments with different parameters as shown in Table 4. The initial trust score of every device is assigned according to the criteria mentioned in Section 3. To show the robustness and efficiency of our method, we conducted experiments to observe the overall influence of different factors and modules such as the influence of weight factors, the influence of observer module, and finally the detection capability of ML-based security module. The empirical results of these experiments are presented as follows:

[6] Car Hacking: Attack & Defense Challenge 2020.

1	Timestamp	Arbitration_ID	DLC	Data	Class	SubClass
2	16.05212	58B	8	11 00 00 00 00 00 00 00	Normal	Normal
3	16.05236	130	8	14 80 10 80 00 00 0A 73	Normal	Normal
4	16.0526	140	8	80 7F 00 73 20 00 0A A1	Normal	Normal
5	16.05342	356	8	00 00 00 80 09 00 00 00	Normal	Normal
6	16.05363	366	7	17 60 0A 17 16 00 01	Normal	Normal
7	16.05388	367	8	00 00 00 00 06 00 2A 0A	Normal	Normal
8	16.05412	368	8	00 00 00 00 00 60 0A 40	Normal	Normal
9	16.05436	38D	8	00 00 49 00 C0 7F FE 79	Normal	Normal
10	16.0546	340	8	00 00 00 24 F6 01 85 30	Normal	Normal

Fig. 2. Snapshot of Car Hacking Data.

Table 3
Snapshot of ToN_IoT Data [44].

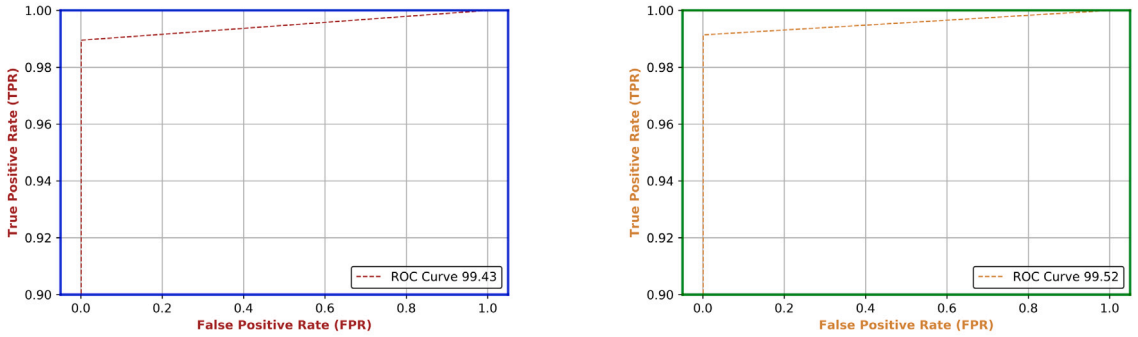
Category	Number of records
Normal data samples	245 000
DDoS attacks data	25 000
Password attacks data	35 000
Scanning attacks data	3973
XSS attacks data	6116
Ransomware attacks data	16 030
Injection attacks data	35 000
Backdoor attacks data	35 000



Fig. 3. Weight factors influence on varying values.

4.3.1. Weight influence in trust assessment

The proposed framework contains important weight factors that can influence the $\mathcal{TA}$ process. These three weight factors (α, β, and γ) are used to confirm the influence of various weight factors on varying sources. As shown in Fig. 3, we conducted experiments to determine their impact on the trustworthiness of the system by setting the values as mentioned in Table 4. Using same α value, it has been observed that the framework offers a low $\mathcal{TS}$ or a high $\mathcal{TS}$ when the weight is $\gamma < \beta$, as compared to $\gamma > \beta$. These same weight factors can be utilized for β, and γ conditions to provide more effective and reliable results. It is worth mentioning that these weight factors have huge impact on the performance of the system. Hence, the $\mathcal{TA}$ process can be effectively tuned to attain desired security level. Furthermore, these weight factors can be utilized properly to assist the prioritization of sources, thus making the proposed framework more efficient, unbiased, and highly reliable.



(a) CHADC Data (b) ToN_IoT Data

Fig. 4. AUC comparison of proposed trust framework using both data.

Table 4
Weight factors values.

Factor	ValueSet1	ValueSet2	ValueSet3
Trust-Threshold	0.60	0.60	0.50
α	0.65	0.65	0.65
β	0.70	0.60	0.60
γ	0.60	0.50	0.60

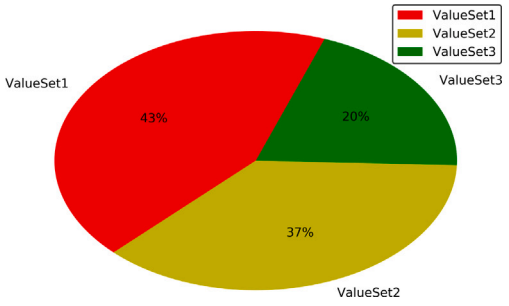


Fig. 5. Weight factors comparison using varying ValueSets.

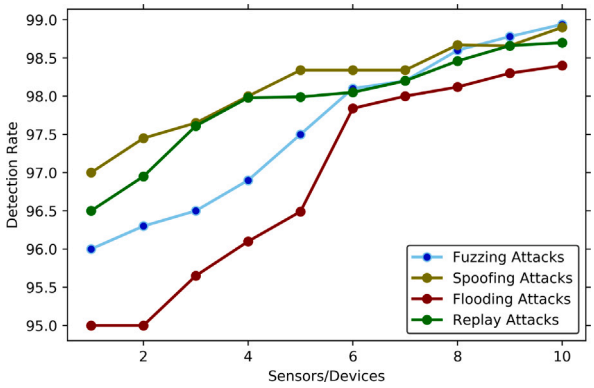


Fig. 6. Detection rates of different types of cyber-attacks.

4.3.2. Observer module influence

The proposed framework contain an observer component to detect malicious nodes in the network. This module is responsible to monitor the activities of the nodes/devices that act maliciously since their $\mathcal{TS}$ is below than the *Trust-Threshold*. As shown in Fig. 5, the proposed framework was able to successfully classifies malicious nodes as non-trustworthy (as their $\mathcal{TS}$ is below than the *Trust-Threshold*). Here, this module also verify impact of weight factors by showing how different values of $\mathcal{TS}$ and weight factors impact the framework. It is evident from these results that a better security performance can be achieved by properly tuning the weight factors. Additionally, due to the context-aware nature of the feedback and feedback crawler mechanisms, the malicious devices can be synced to the entire framework thus helping to determine and observe malicious activities by these nodes.

4.3.3. Attack-detection efficiency and comparative analysis

The effectiveness of the proposed framework is tested using an optimized XGB model and the performance results are outlined in Table 5, using different evaluation metrics such as: Accuracy, F1-measure, Precision, and Recall. As can be seen from the results, the proposed model achieved 99.43% accuracy, 99.42% F1-measure, 99.44% precision, and 98.94% recall rate for CHADC Data. While, the proposed framework achieved 99.52% accuracy, 99.51% F1-measure, 99.53% precision, and 99.10% recall rate for ToN_IoT Data. The detection performance of the proposed framework is also visualized using the ROC[7] curve to verify the robustness of our system. This curve is a key evaluation metric that can be used to plot and summarize the comprehensive accomplishments of model's performance. As seen from Fig. 4, the proposed framework achieved 99.43% and 99.52% area under the curve (for CHADC Data, and ToN_IoT Data, respectively) resulting efficient performance and demonstrating its powerful capability to identify cyber-attacks successfully. The detection rates of different types of cyber-attacks are visualized in Fig. 6, which shows the robustness of proposed framework against evolving attacks with steady detection rate having varied number of devices.

We also compare the performance of proposed framework with other SOTA[8] methods, such as: [16,45–48], and [49]. The empirical results of this comparison are outlined in Table 6 and it can be realized from this table that the proposed framework outperformed all the other comparison methods in terms of better performance and higher accuracy rate showing an average improvement of 5.99% over other benchmark models. It is evident from these results that the proposed framework outperformed SOTA approaches, however this study uses trust tables that are build on historical communication's experience among other sensors which can be exploited by providing malicious trust values for other devices.

In conclusion, the proposed ML-enabled context-aware ZT security framework offers several advantages over other conventional methods by achieving better performance with an average accuracy of 99.47%. Hence, this study proves the ability of ZT architectures in security field. Further, the XGBoost-based security module has proved to have several merits over other ML models such as low error rate, simplified computations, and fast convergence.

[7] Receiver Operating Characteristic.

[8] State-of-the-Art

Table 5
Numerical results of proposed trust framework for both datasets *(A = Accuracy, F1 = F1-score, P = Precision, and R = Recall/Detection rate).

Dataset	A	F1	P	R
CHADC Data	99.43%	99.42%	99.44%	98.94%
ToN_IoT Data	99.52%	99.51%	99.53%	99.1%
Average	**99.47%**	**99.46%**	**99.48%**	**99.03%**

Table 6
Comparison of experimental results of proposed trust-based framework with SOTA methods *(A = Accuracy, SDN = Software Defined Networking).

No	Framework	A(%)	Ref.
1.	Zero Trust IDS	96	[45]
2.	ZT-NIDS	93.01	[46]
3.	Fog-cloud driven ensemble model	96.35	[16]
4.	CNN Model	94.15	[47]
5.	Reinforcement technique combined with SDN method	83.10	[48]
6.	Distributed attack detection	98.27	[49]
	Proposed (average)	**99.47**	
	Average Improvement = 5.99%		

5. Conclusion and future work

Security threats against self-driving or autonomous vehicles are increasing day-by-day which can affect the progression of intelligent transportation systems as well as humans' lives. IDSs offer operational resolves to defend SDVs and their networks from manifold sorts of cyber-threats. This paper proposed a zero trust security framework to guard IoT-based SDVs against complex cyber-attack vectors. The proposed framework integrates contexts of communicating devices and sensors in the trust assessment process. The reliability and effectiveness of the framework is ensured through an observer module and state-of-the-art machine learning algorithm. Through extensive experiments, the proposed framework achieved better performance as compared to benchmark models. In future, we plan to develop federated trust models to provide end-to-end security solution for multiple autonomous vehicle companies for early stage detection of attacks through zero trust approach.

CRediT authorship contribution statement

Izhar Ahmed Khan: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Marwa Keshk:** Writing – review & editing, Methodology, Data curation. **Yasir Hussain:** Software, Methodology, Formal analysis. **Dechang Pi:** Supervision, Funding acquisition. **Bentian Li:** Writing – review & editing, Resources, Conceptualization. **Tanzeela Kousar:** Writing – review & editing, Methodology, Data curation. **Bakht Sher Ali:** Writing – review & editing, Visualization, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used in this study is publically available.

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